



Instituto Tecnológico y de Estudios Superiores de Monterrey

Master of Applied Artificial Intelligence

SCOUT - Supply Chain Operational Vulnerability and Threat Assessment

Project Statement

Team 44

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SCOUT Project Profile

SCOUT - Supply Chain Operational Vulnerability and Threat Assessment is an AI-enabled web application designed to help supply chain professionals translate cybersecurity conditions and technical vulnerabilities into operational priorities. Through a guided self-assessment, the platform will combine technical evidence with supply chain factors such as process criticality, vendor dependence, redundancy, safety implications, recovery capacity, and potential disruption to physical operations. Its intended outputs include a traceable risk profile, prioritized actions, disruption-oriented explanations, and contextual learning guidance.

Institutional Context and Application Setting

The beneficiary institution for SCOUT is the Massachusetts Institute of Technology (MIT), specifically the Supply Chain Cybersecurity Lab within the MIT Center for Transportation and Logistics (MIT CTL). The Lab originated the initiative and will receive the resulting minimum viable product as a research and assessment tool for its work on the operational effects of cyberattacks on physical supply chains. The project will be applied within MIT CTL, located at 1 Amherst Street, Cambridge, Massachusetts 02142, United States.

Under the SCIAN classification, MIT corresponds to Sector 61, Educational Services, and subsector 6113, Higher Education Institutions. More specifically, its organizational profile corresponds most closely to industry 611310, Private Higher Education Institutions. This classification describes the principal activity of the beneficiary institution, while the Supply Chain Cybersecurity Lab identifies the research division in which SCOUT will be applied.

Project Sponsorship and Academic Collaboration

Dr. Camilo A. Mora-Quinones serves as the project sponsor. He is a Postdoctoral Associate at MIT CTL, Founder of the MIT CTL Informal Logistics Initiative, and Co-Founder of the Supply Chain Cybersecurity Lab. In this role, he provides subject-matter guidance and supports the alignment of SCOUT with the Lab's supply chain cybersecurity research. His institutional email address is camimora@mit.edu.

Artificial Intelligence Application Domain

The predominant artificial intelligence application domain selected for SCOUT is Recommendation Systems. The platform is intended to interpret assessment inputs and supporting evidence in order to prioritize operational cyber risks and recommend context-specific actions, including mitigation, recovery, or further analysis. Although SCOUT also incorporates elements of business intelligence and decision support, its principal AI-enabled function is not limited to describing organizational conditions; it transforms those conditions into differentiated recommendations and learning guidance for the user. Authoritative risk calculations will remain transparent and traceable, while AI components may support the contextualization and communication of the resulting recommendations.

Committed Deliverables Plan

The following plan retains all mandatory course deliverables and adapts the intermediate advances to the methodological and technical needs of SCOUT. Each adaptation preserves the academic purpose of the corresponding reference deliverable while producing evidence appropriate to an assessment, recommendation, and decision-support application.

Week	Course reference	SCOUT committed deliverable	Rationale
1	Project statement	Project statement	Mandatory deliverable retained without adaptation.
2	Advance 0: Project proposal and agreements	Project proposal, MVP definition, scope decisions, and required agreements	Mandatory deliverable retained and complemented with the project's validated scope and governance decisions.
3	Advance 1: Exploratory data analysis	Problem, user, evidence-source, and assessment-context analysis	Adapted because SCOUT does not begin with a single modeling dataset. The deliverable will characterize available vulnerability intelligence, organizational inputs, synthetic scenarios, data quality, access constraints, and candidate evaluation evidence.
4	Advance 2: Feature engineering	Assessment instrument, operational risk variables, and initial data model	Preserves the purpose of transforming raw information into usable analytical inputs by defining questionnaire variables, derived factors, unknown-value handling, relationships, and data structures.
5	Advance 3: Baseline	Transparent rules-based prioritization baseline and controlled test cases	Retains the baseline concept through a reproducible reference method against which alternative recommendation or prioritization approaches can be evaluated.
6	Advance 4: Alternative models	Comparative evaluation of alternative prioritization and recommendation approaches	Adapts model comparison to SCOUT by examining candidate methods according to data requirements, explainability, traceability, consistency, and operational usefulness.
7	Advance 5: Final model	Selected methodology and integrated assessment-to-recommendation MVP workflow	Represents the selected analytical approach and its integration with the questionnaire, data layer, risk outputs, recommendations, and basic visualization.
8	Advance 6: Dissemination product	Dissemination product	Mandatory deliverable retained without adaptation.
9	Advance 7: Executive summary	Executive summary	Mandatory deliverable retained without adaptation.
10-11	Final presentation	Final presentation and functional MVP demonstration	Mandatory deliverable retained and supported by the final prototype, evaluation evidence, limitations, and recommendations for future development.

This plan constitutes the team's initial commitment and may be refined with the academic advisor when the project's scope, available evidence, or methodological requirements justify an adjustment.

Project Repository

The shared project repository is available at <https://github.com/Alberto-RodriguezS/SCOUT>. It will be maintained as the controlled workspace for project materials and course evidence.

- **README:** Project purpose, objectives, team members, current scope, and an explanation of the repository structure.
- **Data:** Public or synthetic data used by the project, together with source, structure, access, and data-dictionary information when the data cannot be stored directly.
- **Notebooks:** Analysis, experimentation, scoring prototypes, test cases, and other reproducible development work.
- **Documentation:** Course deliverables, methodological decisions, architecture materials, meeting evidence, and the required week 1, week 2, week 8, and week 9 submissions.

The repository will not contain credentials, passwords, tokens, confidential organizational information, or data that cannot be shared with the team and academic advisor.